

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_random_replay.
- Best CVRPLIB holdout gap: cvrp_random_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_195810_s
- started_at_local: 2026-05-13 19:58:10
- finished_at_local: 2026-05-13 20:01:12
- duration_hhmm: 00:03
- duration_seconds: 181.988
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.247184	0.005376	0.139714	0.925859	0.58	-0.003142
cvrp_random_replay	random	score_only	False	0.222852	0.00754	0.127158	0.795003	0.78	0.012852
cvrp_failure_replay	failure	score_only	False	0.246867	0.005376	0.139538	0.917671	0.58	-0.004132
cvrp_failure_replay_compression	failure	novelty_gate	True	0.260052	0.005376	0.146863	0.762695	0.58	0.036652

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.247184, synthetic 0.005376, combined 0.139714.

- Accepted-epoch count 3, mean accepted code novelty 0.925859, and final complexity 0.58.
- Adaptation efficiency -0.003142 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.247184 across 5 instances; family means: A=0.138743, B=0.254873, E=0.285988, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 587, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.281659}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.222852, synthetic 0.00754, combined 0.127158.
- Accepted-epoch count 2, mean accepted code novelty 0.795003, and final complexity 0.78.
- Adaptation efficiency 0.012852 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.222852 across 5 instances; family means: A=0.120419, B=0.228589, E=0.28215, P=0.254513.
- Panel synthetic_holdout mean gap 0.00754 across 4 instances; family means: alternating_belt=0.03016, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 458, "cost": 574, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.253275}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.246867, synthetic 0.005376, combined 0.139538.
- Accepted-epoch count 2, mean accepted code novelty 0.917671, and final complexity 0.58.
- Adaptation efficiency -0.004132 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.246867 across 5 instances; family means: A=0.103839, B=0.267694, E=0.293666, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.260052, synthetic 0.005376, combined 0.146863.
- Accepted-epoch count 2, mean accepted code novelty 0.762695, and final complexity 0.58.
- Adaptation efficiency 0.036652 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.260052 across 5 instances; family means: A=0.103839, B=0.267694, E=0.339731, P=0.3213.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 602, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.31441}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of Replay-aware CVRP Benchmark Results

Condition	Replay Mode	Held-out CVRP LIB Gap	Synthetic Holdout Gap	Transfer Gap	Adaptation Efficiency	Code Novelty (Mean)	Complexity	Notes on Trends
cvrp_no_replay	None	0.2472	**0.00538**	0.1397	-0.0031	**0.926**	0.58	Highest code novelty, moderate transfer gaps
cvrp_random_replay	Random Replay	**0.2229 (best)**	0.00754	**0.1272 (best)**	0.0129	0.795	0.78	Best transfer and heldout gaps, lower code novelty vs no replay, higher complexity
cvrp_failure_replay	Failure Replay	0.2469	0.00538	0.1395	-0.0041	0.918	0.58	Transfer gaps similar to no replay, code novelty similar to no replay
cvrp_failure_replay_compression	Failure Replay + Compression	0.2601	0.00538	0.1469	0.0367	0.763	0.58	Worst CVRPLIB gap and transfer, lowest code novelty, positive adaptation efficiency

Interpretation:

- **Main transfer evidence (Held-out CVRPLIB and Synthetic gaps)**:
- **Random replay condition (cvrp_random_replay) shows the best transfer performance**, with lowest mean CVRPLIB gap (0.2229) and lowest mean transfer gap (0.1272). It also shows higher synthetic holdout gap (0.00754) than no replay but still small.
- **No replay (cvrp_no_replay) achieves the best synthetic holdout gap (0.00538)** but has worse heldout CVRPLIB and transfer gaps than random replay.
- Failure replay conditions (cvrp_failure_replay, cvrp_failure_replay_compression) do not improve transfer gaps compared to no replay and generally are worse on CVRPLIB gap, especially with compression-aware mode.
- **Code novelty vs. transfer trade-off**:

- The no replay condition has the highest code novelty (~0.926) but lower transfer performance.
- Random replay reduces code novelty (~0.795) but improves transfer metrics substantially. This suggests a trade-off where replay degrades novelty but improves solution quality on held-out and transferred problems.
- Failure replay conditions have relatively high code novelty (~0.918, 0.763) but do not improve transfer compared to no replay.
- **Complexity**:
 - Random replay yields higher complexity (0.78) vs no replay and failure replay (both ~0.58). This may relate to more complex behavior profiles enabling better transfer.
 - Compression-aware failure replay lowers code novelty and increases adaptation efficiency positively but degrades transfer and optimality gaps.
- **Adaptation efficiency**:
 - Only random replay and compression-aware failure replay show positive adaptation efficiency, indicating efficiency in learning or adapting new solutions.
 - Despite positive efficiency, compression-aware failure replay has worse gaps, highlighting efficiency alone is insufficient for transfer quality.

Conservative Conclusions:

- **Random replay delivers the best transfer performance on held-out CVRPLIB and synthetic problems.** This is the strongest evidence for beneficial replay use in the benchmark.
- **No replay attains the best synthetic holdout gap and highest code novelty, but worse overall transfer and heldout CVRPLIB gaps than random replay.** Novelty decreases under random replay, indicating replay may reduce code diversity but improve solution generalization.
- **Failure replay modes do not improve transfer gaps compared to no replay and may degrade performance especially with compression pressure despite some gains in adaptation efficiency or complexity.**
- **Increased lexical or code novelty does not correlate with improved transfer or held-out optimality gaps. Thus, code novelty is not a proxy for algorithmic invention in this context.**
- **Random replay has the most favorable balance of transfer gaps and adaptation efficiency, but at the cost of increased solution complexity and decreased code novelty.**

Final Recommendation:

Prioritize **random replay** for transfer quality improvements in replay-aware CVRP benchmarks. Avoid failure replay and overemphasis on code novelty or compression-aware replay alone, as these do not translate to better transfer or heldout gaps.